

AI Feasibility Assessment: ****Highly Recommended****

While core banking (transactions, account management) should remain deterministic and rule-based, AI is critical for the "value-added" features requested (Financial Insights) and essential for modernizing Risk Management (Fraud Detection).

1. AI Overview

- ****Purpose:**** To transform historical transactional data into predictive insights and real-time security intelligence.
- ****Key Modules:****
 - * **Financial Wellness Engine:** Personalized budgeting and spending categorization.
 - * **Fraud Detection Engine:** Anomaly detection on transaction streams.
 - * **Intelligent Underwriting Assistant:** Summarizing applicant data to accelerate human decision-making.

2. Models

- ****Predictive Modeling (Fraud):**** XGBoost or LightGBM for tabular transaction data. These are faster, explainable, and highly effective for binary classification (Fraud/Not Fraud).
- ****Generative AI (Insights/Assistant):**** GPT-4o or Claude 3.5 Sonnet (via API) for natural language processing, generating human-readable financial advice and summarizing loan application dossiers.
- ****Clustering (Categorization):**** K-Means or HDBSCAN for grouping merchant data into categories (Groceries, Utilities, Entertainment).

3. Prompt Engineering

- ****Persona:**** "You are a Senior Financial Advisor for a retail bank."
- ****System Prompt:**** "Analyze the user's spending data for the past 30 days. Identify outliers and provide actionable, empathetic advice for budget optimization. Maintain a professional, non-judgmental tone. Never provide specific investment advice."
- ****Technique:**** Few-shot prompting (providing examples of "Good" vs "Bad" financial advice) to ensure the tone aligns with banking regulations.

4. RAG (Retrieval-Augmented Generation) Design

- ****Context:**** Use RAG to ingest internal bank policy documents and the user's specific transactional history.

- **Workflow:** When a user asks "Can I afford this loan?", the system retrieves (1) the user's latest balance/spending history from the DB and (2) the bank's "Loan Eligibility Policy" PDF. The LLM then synthesizes these to provide an informed answer.

5. Memory

- **Short-term:** Maintain conversation state (Redis) to allow follow-up questions about specific budget items.
- **Long-term:** Store user preferences and "Insights History" in a SQL/NoSQL database to ensure the AI remembers past advice given to the customer.

6. Agents

- **The Insights Agent:** Orchestrates access to the transaction database, runs analytics, and formats the output for the UI.
- **The Fraud Analyst Agent:** Monitors transaction streams; if it triggers an alert, it gathers "Evidence" (recent IP addresses, merchant locations) and presents a summary to the human Compliance Officer.

7. Vector Database

- **Choice:** Pinecone or Milvus.
- **Use Case:** Storing embeddings of customer financial behaviors and policy documents to perform semantic search, allowing the system to find similar patterns of successful vs. delinquent borrowers.

8. Embeddings

- **Model:** `text-embedding-3-small` (OpenAI) or `bge-large-en-v1.5`.
- **Application:** Converting unstructured text (Loan application notes, customer feedback) into vectors to enable efficient searching and similarity matching.

9. LLM Workflow

1. **Ingestion:** Transaction occurs.
2. **Feature Extraction:** Anomaly detection flags the event.
3. **Retrieval:** RAG pulls user profile and historical patterns.
4. **Generation:** LLM generates a summary for the human analyst (in fraud cases) or a tip for the user (in insights cases).
5. **Human-in-the-Loop (HITL):** Analyst reviews AI-flagged fraud before any account freeze occurs.

10. AI Risks & Mitigation

- **Hallucination:** LLMs might miscalculate budget projections. **Mitigation:** Force the LLM to use "Tool Use/Function Calling" to fetch calculated values from a Python script rather than calculating them itself.
- **Bias in Lending:** Models might inadvertently discriminate. **Mitigation:** Audit training data for sensitive features (race, gender) and implement "Fairness-Aware" machine learning constraints.
- **Data Privacy:** Sensitive financial data in the cloud. **Mitigation:** Use PII-redaction layers before sending data to any LLM APIs; consider hosting local open-source models (Llama 3) for highly sensitive underwriting summaries.